

Objective: Review of CI/CD pipeline implementations, providing feedback for improvement, Troubleshooting CI/CD issues.

Required Software / Tools: VSCode, GITHUB.

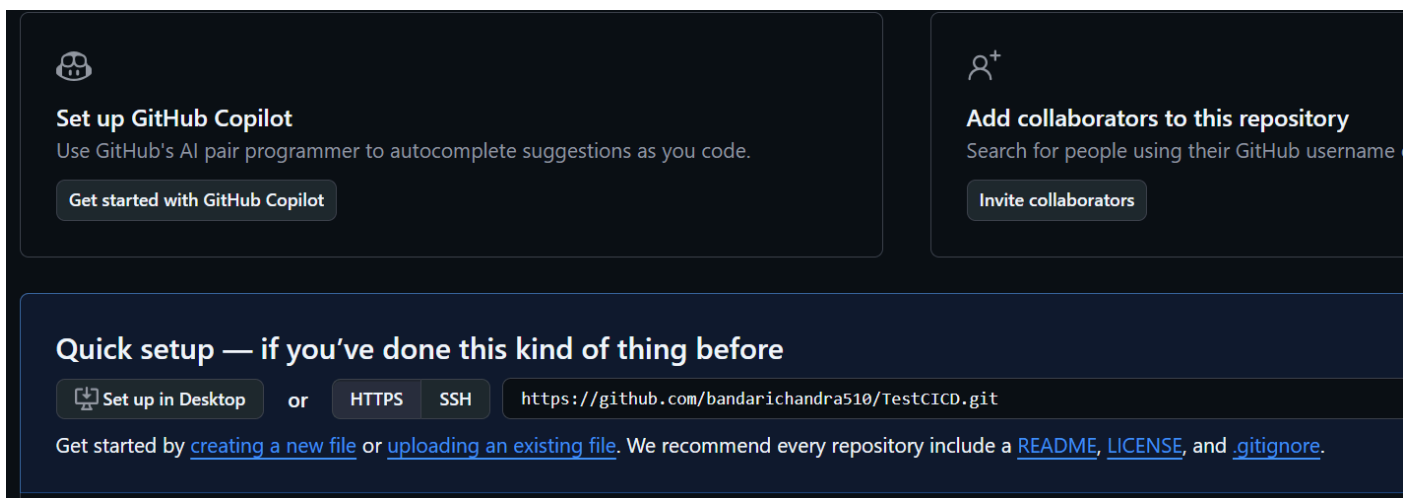
Project Structure:

Exp6/

```
| — app.py
| — test_app.py
| — requirements.txt
| — .github/
|   └─ workflows/
|       └─ ci-cd.yml
```

Procedure:

1. Login to GITHUB
2. Create a new repo with name Exp6.
3. Click create file under repo.



4. Name the file as app.py and paste below code

```
def add(a, b):
    return a + b
```

```
if __name__ == "__main__": # pragma: no cover
    result = add(2, 3)
    print(result)
```

5. Save and Commit changes to main branch.
6. Now create one more file by click add file.
7. Name it as test_app.py and paste below code.

```
from app import add
```

```
def test_add_positive():  
    assert add(2, 3) == 5
```

```
def test_add_negative():  
    assert add(-1, -1) == -2
```

```
def test_add_zero():  
    assert add(0, 5) == 5
```

```
def test_add_mixed():  
    assert add(-1, 1) == 0
```

```
def test_output_type():  
    result = add(10, 5)  
    assert isinstance(result, int)  
    assert result == 15
```

8. Save and Commit changes to main branch.
9. Create new file with name requirements.txt and add below contents

```
pytest
```

```
pytest-cov
```

```
flake8
```

10. Now go to github actions and click “set up a workflow yourself ”
11. Name it ci-cd.yml. Paste below code and save & commit the workflow to main branch.

```
name: Python CI/CD Pipeline
```

```
on:
```

```
  push:
```

```
    branches: ["main"]
```

```
  pull_request:
```

```
    branches: ["main"]
```

```
jobs:
```

```
  build-test:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout code
```

```
        uses: actions/checkout@v4
```

```
      - name: Set up Python
```

```
        uses: actions/setup-python@v5
```

with:

python-version: "3.10"

- name: Install dependencies

run: |

python -m pip install --upgrade pip

pip install -r requirements.txt

- name: Lint code (Quality Check)

run: |

flake8 .

- name: Run tests with coverage

run: |

pytest --cov=app --cov-report=term --cov-fail-under=80

- name: Validate script output

run: |

output=\$(python app.py)

if ["\$output" != "5"]; then

echo "Unexpected output: \$output"

exit 1

fi

deploy:

needs: build-test

runs-on: ubuntu-latest

if: github.ref == 'refs/heads/main'

steps:

- name: Deploy Application

run: echo "Deploying application..."

- name: Post-deployment validation

run: echo "Deployment successful and verified!"

12. Go to Github actions, check pipeline status, fix any failures.

bandarichandra510 / Exp6

Q Type / to search

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<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

← Python CI/CD Pipeline

Update app.py #8

Re-run all jobs

Summary

All jobs

build-test

deploy

Run details

Usage

Workflow file

Triggered via push 1 minute ago

Status

Total duration

Artifacts

bandarichandra510 pushed

8b9c083

main

Success

31s

—

ci-cd.yml

on: push

build-test

14s

deploy

4s

—

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